

Linear, Hamming, and Cyclic Codes

– A Mathematician's Introduction –

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수학의 즐거움, *Enjoying Math*

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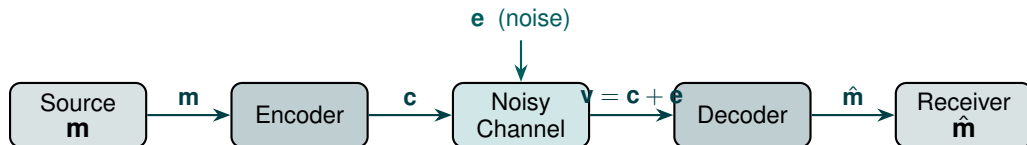
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I. Introduction

I. Introduction

The Communication Problem



Setup

- ▶ Message $\mathbf{m} \in \mathbb{F}_q^k$
- ▶ Encoder adds redundancy: $\mathbf{m} \mapsto \mathbf{c} \in \mathbb{F}_q^n$, $n > k$
- ▶ Channel corrupts: $\mathbf{v} = \mathbf{c} + \mathbf{e}$

Goal

- ▶ Recover $\hat{\mathbf{m}} = \mathbf{m}$ even when $\mathbf{e} \neq \mathbf{0}$
- ▶ Keep rate $R = k/n$ as large as possible

I. Introduction

Shannon's Channel Coding Theorem (1948)



Shannon's Channel Coding Theorem

For any discrete memoryless channel with capacity C and any rate $R < C$, there exists a sequence of codes of rate $\rightarrow R$ whose block error probability $\rightarrow 0$.

What it guarantees

- ▶ Reliable communication *is* possible below capacity
- ▶ Error probability can be made arbitrarily small

What it does not provide

- ▶ *How* to construct such codes explicitly
- ▶ *How* to decode efficiently

The Central Challenge of Coding Theory

Construct **explicit algebraic codes** that are both **efficient** (high rate) and **powerful** (correct many errors).

I. Introduction

Roadmap

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II. Linear Codes

Structure via **linear algebra**

- Subspaces of $\mathbb{F}_q^n$
- Generator G , parity-check H
- Hamming distance & error correction

III. Hamming Codes

Perfect codes via **projective geometry**

- Parity checks from projective points
- Syndrome decoding as geometry
- Perfect single-error correction

IV. Cyclic Codes

Structure via **polynomial rings**

- Quotient $\mathbb{F}_q[X]/(X^n - 1)$
- Ideals and generator polynomials
- BCH design and cyclic Hamming codes

Key Observation

Coding theory is a meeting point of **linear algebra**, **finite geometry**, and **polynomial algebra**.

II. Linear Codes

II. Linear Codes

The ambient space



Hamming space

Fix a finite field $\mathbb{F}_q$. The set $\mathbb{F}_q^n$ is simultaneously a vector space and a metric space with

$$d(x, y) = \#\{i : x_i \neq y_i\}.$$

- ▶ The metric is translation invariant: $d(x + z, y + z) = d(x, y)$.
- ▶ The spheres are combinatorial objects:

$$|B_t(0)| = \sum_{i=0}^t \binom{n}{i} (q-1)^i.$$

- ▶ Coding theory asks for large subsets with large mutual distance.

II. Linear Codes

Codes as metric subsets



Code

A code of length n over $\mathbb{F}_q$ is a subset $\mathcal{C} \subseteq \mathbb{F}_q^n$. Its minimum distance is

$$d(\mathcal{C}) = \min\{d(x, y) : x, y \in \mathcal{C}, x \neq y\}.$$

Mathematician's viewpoint

The fundamental problem is an extremal problem in a finite metric space:

$$A_q(n, d) = \max\{|\mathcal{C}| : \mathcal{C} \subseteq \mathbb{F}_q^n, d(\mathcal{C}) \geq d\}.$$

II. Linear Codes

Linearity



Linear code

A linear $[n, k, d]_q$ code is a k -dimensional subspace

$$\mathcal{C} \leq \mathbb{F}_q^n$$

with minimum distance d .

- ▶ The cardinality is forced: $|\mathcal{C}| = q^k$.
- ▶ The rate is $R = k/n$.
- ▶ Since $\mathcal{C}$ is a group under addition,

$$d(\mathcal{C}) = \min\{\text{wt}(c) : 0 \neq c \in \mathcal{C}\}.$$

II. Linear Codes

Weight



Hamming weight

For $x = (x_1, \dots, x_n) \in \mathbb{F}_q^n$,

$$\text{wt}(x) = |\text{supp}(x)|, \quad \text{supp}(x) = \{i : x_i \neq 0\}.$$

Lemma

For a linear code $\mathcal{C}$,

$$d(\mathcal{C}) = \min_{0 \neq c \in \mathcal{C}} \text{wt}(c).$$

Proof. If $x, y \in \mathcal{C}$, then $x - y \in \mathcal{C}$ and $d(x, y) = \text{wt}(x - y)$.

II. Linear Codes

Generator matrices

Generator matrix

A generator matrix for an $[n, k]_q$ code is a matrix $G \in \mathbb{F}_q^{k \times n}$ whose rows form a basis of $\mathcal{C}$. Encoding is the linear map

$$\mathbb{F}_q^k \longrightarrow \mathbb{F}_q^n, \quad m \longmapsto mG.$$

- ▶ The code is the row space of G .
- ▶ Different bases give different generator matrices for the same code.
- ▶ Replacing G by UG , $U \in \text{GL}_k(\mathbb{F}_q)$, changes coordinates on messages only.

II. Linear Codes

Equivalence of descriptions



Row-space equivalence

Two full-rank matrices $G, G' \in \mathbb{F}_q^{k \times n}$ generate the same code if and only if

$$G' = UG$$

for some $U \in \text{GL}_k(\mathbb{F}_q)$.

- ▶ This is just change of basis in the subspace $\mathcal{C}$.
- ▶ The code is the invariant object; the matrix is a coordinate choice.
- ▶ Many computational algorithms are Gaussian elimination on this statement.

II. Linear Codes

Systematic form



Systematic generator matrix

After permuting coordinates, many generator matrices can be written

$$G = [I_k \mid P].$$

- ▶ The message appears visibly in the first k coordinates.
- ▶ The remaining $n - k$ coordinates are parity symbols.
- ▶ Systematic form is convenient, but it is not an intrinsic property of the code.

Caution

Coordinate permutations preserve distance, but they change the visible positions of message and parity symbols.

II. Linear Codes

Parity-check matrices



Parity-check matrix

A parity-check matrix is a matrix $H \in \mathbb{F}_q^{(n-k) \times n}$ such that

$$\mathcal{C} = \ker(H) = \{c \in \mathbb{F}_q^n : Hc^T = 0\}.$$

- ▶ If G generates $\mathcal{C}$, then $GH^T = 0$.
- ▶ The rows of H generate the dual code $\mathcal{C}^\perp$.
- ▶ The equation $Hc^T = 0$ is a system of homogeneous parity constraints.

II. Linear Codes

Duality



Dual code

The dual of $\mathcal{C} \leq \mathbb{F}_q^n$ is

$$\mathcal{C}^\perp = \{y \in \mathbb{F}_q^n : \langle x, y \rangle = 0 \text{ for all } x \in \mathcal{C}\}.$$

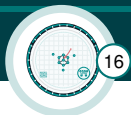
Dimension identity

$$\dim \mathcal{C} + \dim \mathcal{C}^\perp = n.$$

- ▶ A parity-check matrix for $\mathcal{C}$ is a generator matrix for $\mathcal{C}^\perp$.
- ▶ Duality exchanges constraints and codewords.

II. Linear Codes

Distance from checks



Column criterion

Let H be a parity-check matrix of $\mathcal{C}$. Then $d(\mathcal{C})$ is the least number of columns of H that are linearly dependent.

Proof.

A vector c of weight s satisfies $Hc^T = 0$ exactly when a linear combination of the s corresponding columns is zero. □

- ▶ No zero column means $d \geq 2$.
- ▶ No proportional columns means $d \geq 3$.
- ▶ This criterion is the entrance to Hamming codes.

II. Linear Codes

Syndromes



Syndrome

For a received word $v = c + e$, define

$$s(v) = Hv^T.$$

Since $Hc^T = 0$, one has

$$s(v) = He^T.$$

- ▶ The syndrome depends only on the error coset.
- ▶ $s(v) = 0$ if and only if $v \in \mathcal{C}$.
- ▶ Decoding is the problem of choosing a likely error in each coset.

II. Linear Codes

Cosets



Coset decomposition

The ambient space is a disjoint union of cosets:

$$\mathbb{F}_q^n = \coprod_{a \in A} (a + \mathcal{C}),$$

where $|A| = q^{n-k}$.

- ▶ All words in the same coset have the same syndrome.
- ▶ Syndromes parameterize cosets when H has full rank.
- ▶ Decoding assigns to each coset a chosen representative of small weight.

II. Linear Codes

Standard arrays



Coset leader

A coset leader is an element of least Hamming weight in a coset $a + \mathcal{C}$.

- ▶ The standard array lists cosets by their leaders.
- ▶ Nearest-neighbor decoding subtracts the coset leader.
- ▶ The method is conceptually clean but exponential in $n - k$.

Principle

Good algebraic families replace the standard array by structure.

II. Linear Codes

Correction and detection

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Distance theorem

A code of minimum distance d detects every error of weight at most $d - 1$ and corrects every error of weight at most

$$t = \left\lfloor \frac{d-1}{2} \right\rfloor.$$

- ▶ Detection: no nonzero error of weight $< d$ is a codeword.
- ▶ Correction: balls of radius t around distinct codewords are disjoint.

II. Linear Codes

Packing

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Hamming bound

If $\mathcal{C}$ is an $[n, k, d]_q$ code and $t = \lfloor (d-1)/2 \rfloor$, then

$$q^k \sum_{i=0}^t \binom{n}{i} (q-1)^i \leq q^n.$$

Proof.

The balls of radius t around codewords are disjoint subsets of $\mathbb{F}_q^n$. □

Perfect code

Equality means the radius- t balls tile the whole Hamming space.

II. Linear Codes

Singleton bound



Singleton bound

Every $[n, k, d]_q$ code satisfies

$$d \leq n - k + 1.$$

Proof.

Delete $d - 1$ coordinates. Distinct codewords must remain distinct, so $q^k \leq q^{n-d+1}$. □

- ▶ Codes meeting the bound are called MDS codes.
- ▶ Reed–Solomon codes are the standard nonbinary MDS examples.

II. Linear Codes

Gilbert–Varshamov idea

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Existence principle

There exist linear codes whose parameters are good because a greedy construction cannot cover the whole space too early.

$$q^k \sum_{i=0}^{d-2} \binom{n-1}{i} (q-1)^i < q^n \quad \Rightarrow \quad \text{an } [n, k, d]_q \text{ code exists.}$$

- ▶ This is not a construction one wants to implement.
- ▶ It proves that the asymptotic landscape is nontrivial.
- ▶ Algebraic families try to approach such existence bounds explicitly.

II. Linear Codes

Griesmer bound

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Griesmer bound

For a linear $[n, k, d]_q$ code,

$$n \geq \sum_{i=0}^{k-1} \left\lceil \frac{d}{q^i} \right\rceil.$$

- ▶ This is stronger than Singleton in many small-parameter cases.
- ▶ It comes from repeatedly puncturing on the support of a minimum word.
- ▶ It is a linear-code bound: linearity is essential.

II. Linear Codes

Puncturing



Puncturing

Puncturing a code at coordinate i means deleting the i -th coordinate from every word.

- ▶ Length decreases by one.
- ▶ Dimension usually stays the same, but can drop.
- ▶ Minimum distance can decrease by at most one.

Use

Puncturing is the basic operation for inductive arguments on code parameters.

II. Linear Codes

Shortening



Shortening

Shortening at coordinate i means first selecting codewords with $c_i = 0$, then puncturing at i .

- ▶ Length decreases by one.
- ▶ Dimension can decrease by one.
- ▶ Minimum distance does not decrease.

Duality

Puncturing and shortening are dual operations:

$$(\text{puncture } \mathcal{C})^\perp = \text{shorten } (\mathcal{C}^\perp).$$

II. Linear Codes

Weight enumerators



Weight enumerator

The weight enumerator is

$$W_C(X, Y) = \sum_{c \in C} X^{n-\text{wt}(c)} Y^{\text{wt}(c)}.$$

- ▶ It records the distribution of weights.
- ▶ The minimum distance is the first nonzero exponent of Y after the constant term.
- ▶ It is a coarser invariant than the code but a powerful one.

II. Linear Codes

MacWilliams identity



MacWilliams identity

For a linear $[n, k]_q$ code,

$$w_{\mathcal{C}^\perp}(X, Y) = \frac{1}{|\mathcal{C}|} w_{\mathcal{C}}(X + (q-1)Y, X - Y).$$

- ▶ The identity is a finite Fourier transform.
- ▶ It turns the weight distribution of $\mathcal{C}$ into that of $\mathcal{C}^\perp$.
- ▶ It is one of the central reasons duality is computationally useful.

II. Linear Codes

Self-orthogonality



Self-orthogonal and self-dual

A code is self-orthogonal if $\mathcal{C} \subseteq \mathcal{C}^\perp$ and self-dual if $\mathcal{C} = \mathcal{C}^\perp$.

- ▶ Self-dual binary codes have dimension $n/2$.
- ▶ Doubly-even self-dual codes have all weights divisible by 4.
- ▶ Such codes connect coding theory to lattices and modular forms.

II. Linear Codes

Equivalent codes



Monomial equivalence

Two q -ary linear codes are equivalent if one is obtained from the other by permuting coordinates and multiplying coordinates by nonzero scalars.

- ▶ Over $\mathbb{F}_2$, this is just coordinate permutation.
- ▶ Equivalent codes have the same parameters and weight enumerator.
- ▶ Classification means classification up to this equivalence.

II. Linear Codes

Automorphisms



Automorphism group

$$\text{Aut}(\mathcal{C}) = \{\pi \in S_n : \pi(\mathcal{C}) = \mathcal{C}\}$$

in the binary case.

- ▶ A large automorphism group is a sign of hidden geometry.
- ▶ Symmetry reduces decoding complexity.
- ▶ Hamming and cyclic codes both arise from strong symmetry.

II. Linear Codes

Repetition code

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Repetition code

$$\mathcal{C} = \{(a, a, \dots, a) : a \in \mathbb{F}_q\}$$

is an $[n, 1, n]_q$ code.

- ▶ It has maximal distance for dimension 1.
- ▶ Its rate is $1/n$, so it is inefficient.
- ▶ It corrects $\lfloor (n-1)/2 \rfloor$ errors by majority vote when $q = 2$.

II. Linear Codes

Single parity-check code



Parity-check code

$$\mathcal{C} = \{x \in \mathbb{F}_q^n : x_1 + \cdots + x_n = 0\}$$

is an $[n, n-1, 2]_q$ code.

- ▶ The parity-check matrix is $H = [1 \ 1 \ \cdots \ 1]$.
- ▶ It detects one error but cannot correct one error.
- ▶ It is the dual of the repetition code.

II. Linear Codes

The simplex code



Binary simplex code

Let H_r be the matrix whose columns are all nonzero vectors of $\mathbb{F}_2^r$. The row space of H_r is the simplex code.

- ▶ It has parameters $[2^r - 1, r, 2^{r-1}]_2$.
- ▶ Every nonzero codeword has the same weight.
- ▶ Its dual is the binary Hamming code.

II. Linear Codes

Matroid viewpoint



Vector matroid of a code

A parity-check matrix H defines a matroid on the coordinate set $\{1, \dots, n\}$ by linear dependence of columns.

- ▶ Circuits of the matroid correspond to minimal support codewords.
- ▶ The minimum distance is the size of the smallest circuit.
- ▶ Projective geometry enters when the columns of H are projective points.

II. Linear Codes

Tanner graphs



Tanner graph

Given a parity-check matrix H , form a bipartite graph with variable nodes $1, \dots, n$ and check nodes equal to the rows of H .

- ▶ An edge joins variable j to check i when $H_{ij} \neq 0$.
- ▶ Sparse H gives iterative decoding algorithms.
- ▶ Classical algebraic codes often have dense checks; LDPC codes emphasize sparsity.

II. Linear Codes

Decoding hardness

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Syndrome decoding problem

Given H , a syndrome s , and an integer t , decide whether there exists e with

$$He^T = s, \quad \text{wt}(e) \leq t.$$

- ▶ This problem is NP-complete in general.
- ▶ Code-based cryptography is built around this hardness.
- ▶ Algebraic code families are useful because they add trapdoor-like structure.

II. Linear Codes

Perfect codes

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Perfect code

A code correcting t errors is perfect if the radius- t balls around codewords partition $\mathbb{F}_q^n$.

$$|\mathcal{C}| \sum_{i=0}^t \binom{n}{i} (q-1)^i = q^n.$$

- ▶ Perfect codes are extremely rare.
- ▶ The binary Hamming codes are perfect one-error-correcting codes.
- ▶ The Golay codes are the exceptional perfect codes.

II. Linear Codes

Why linear codes are central



Algebra

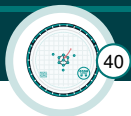
- ▶ subspaces
- ▶ duality
- ▶ matrices
- ▶ finite fields

Algorithms

- ▶ encoding by G
- ▶ checking by H
- ▶ syndrome decoding
- ▶ structured families

II. Linear Codes

The road to Hamming



Question

Can we construct a linear code that corrects one error and wastes no syndrome?

- ▶ One error at one of n positions gives n possibilities.
- ▶ No error gives one more possibility.
- ▶ A parity-check matrix with r rows has 2^r binary syndromes.

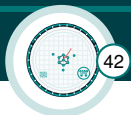
$$n + 1 = 2^r$$

is the numerology of binary Hamming codes.

III. Hamming Codes

III. Hamming Codes

The design problem



Goal

Construct binary linear codes that correct exactly one error and use all possible syndromes.

- ▶ A received word has either no error or one error in position i .
- ▶ Thus there are $1 + n$ elementary cases.
- ▶ With r parity checks, there are 2^r syndromes.

$$1 + n = 2^r \quad \Rightarrow \quad n = 2^r - 1.$$

III. Hamming Codes

The parity-check matrix

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Binary Hamming matrix

For $r \geq 2$, let H_r be the $r \times (2^r - 1)$ matrix whose columns are all nonzero vectors of $\mathbb{F}_2^r$.

$$H_3 = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

- ▶ Column labels are nonzero binary syndromes.
- ▶ Every nonzero syndrome identifies a unique column.

III. Hamming Codes

Definition



Binary Hamming code

The binary Hamming code of order r is

$$\text{Ham}(r, 2) = \ker(H_r) \leq \mathbb{F}_2^{2^r - 1}.$$

- ▶ Length: $n = 2^r - 1$.
- ▶ Redundancy: r .
- ▶ Dimension: $k = n - r = 2^r - 1 - r$.

$\text{Ham}(r, 2)$ has parameters $[2^r - 1, 2^r - 1 - r, 3]_2$.

III. Hamming Codes

Dimension



Dimension count

The matrix H_r has rank r , hence

$$\dim \ker(H_r) = 2^r - 1 - r.$$

Proof.

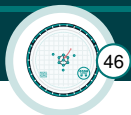
The standard basis vectors $e_1, \dots, e_r$ occur among the nonzero columns of H_r , so the row rank is r . □

Interpretation

The parity checks are independent; no check is redundant.

III. Hamming Codes

Minimum distance



Distance

The binary Hamming code $\text{Ham}(r, 2)$ has minimum distance 3.

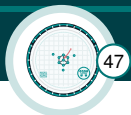
Proof.

No column of H_r is zero, so $d \geq 2$. No two columns are equal over $\mathbb{F}_2$, so no two are dependent, hence $d \geq 3$. But $u + v + (u + v) = 0$, so three suitable columns are dependent. $\square$

- ▶ Therefore Hamming codes correct one error.
- ▶ They detect two errors.

III. Hamming Codes

Perfectness



Hamming codes are perfect

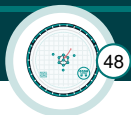
The binary Hamming code satisfies equality in the Hamming bound for $t = 1$.

$$2^k(1 + n) = 2^{2^r - 1 - r}(1 + 2^r - 1) = 2^{2^r - 1} = 2^n.$$

- ▶ Every word of $\mathbb{F}_2^n$ is at distance at most 1 from exactly one codeword.
- ▶ The syndrome is not merely a diagnostic: it is a complete decoder.

III. Hamming Codes

Syndrome decoding



Decoder

Given $v \in \mathbb{F}_2^n$, compute $s = H_r v^T$.

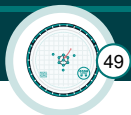
- ▶ If $s = 0$, accept v .
- ▶ If $s \neq 0$, find the unique column $H_i = s$.
- ▶ Correct by flipping coordinate i .

Reason

For a single error e_i , the syndrome is $H_r e_i^T = H_i$.

III. Hamming Codes

Binary index convention



Indexed columns

If the i -th column of H_r is the binary expansion of i , then the nonzero syndrome is literally the error position.

$$s = (1, 0, 1)^T \rightsquigarrow i = 5$$

depending on the chosen bit order.

- ▶ This is a notational convention, not an invariant.
- ▶ It explains the classical parity-bit positions $1, 2, 4, \dots$

III. Hamming Codes

The classical $[7,4,3]$ code



$$H = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

- ▶ There are 7 nonzero syndromes.
- ▶ Each syndrome names one error coordinate.
- ▶ The kernel has dimension 4.

$$\text{Ham}(3,2) = [7,4,3]_2.$$

III. Hamming Codes

A systematic generator matrix



One systematic form for the $[7, 4, 3]$ Hamming code is

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

- ▶ The first four coordinates are message symbols.
- ▶ The last three are parity symbols.
- ▶ This G is equivalent to the kernel of a suitable H .

III. Hamming Codes

Encoding example



For $m = (1, 0, 1, 1)$ and the systematic matrix G ,

$$c = mG = (1, 0, 1, 1, 0, 0, 1).$$

- ▶ The parity bits are linear forms in the message bits.
- ▶ A single flipped bit changes the syndrome to the corresponding column.
- ▶ The correction rule does not require knowing the transmitted message.

III. Hamming Codes

Projective geometry



Projective interpretation

The nonzero columns of H_r , modulo scalar multiplication, are the points of projective space

$$\mathbb{P}^{r-1}(\mathbb{F}_q).$$

- ▶ Over $\mathbb{F}_2$, each nonzero vector is its own projective point.
- ▶ Over $\mathbb{F}_q$, scalar multiples define the same projective point.
- ▶ Hamming codes are parity checks indexed by projective points.

III. Hamming Codes

The Fano plane



For $r = 3$, the projective space $\mathbb{P}^2(\mathbb{F}_2)$ has 7 points and 7 lines.

- ▶ The 7 columns of H_3 are the points.
- ▶ A dependent triple of columns is a line of the Fano plane.
- ▶ Weight-3 codewords are exactly these lines.

Geometry

The minimum distance 3 is the statement that the smallest projective line has three points.

III. Hamming Codes

Lines give codewords



In $\mathbb{P}^2(\mathbb{F}_2)$, if a, b, c are the three points on a line, then

$$a + b + c = 0.$$

- ▶ The incidence vector of that line is a Hamming codeword.
- ▶ The 7 lines give the 7 words of weight 3.
- ▶ Their complements give 7 words of weight 4.

$$W_{\text{Ham}(3,2)}(Y) = 1 + 7Y^3 + 7Y^4 + Y^7.$$

III. Hamming Codes

The simplex dual



Simplex code

The row space of H_r is the binary simplex code:

$$\text{Sim}(r, 2) = \text{rowspan}(H_r).$$

- ▶ It has parameters $[2^r - 1, r, 2^{r-1}]_2$.
- ▶ Every nonzero word has weight 2^{r-1} .
- ▶ It is dual to the Hamming code.

$$\text{Ham}(r, 2)^\perp = \text{Sim}(r, 2).$$

III. Hamming Codes

MacWilliams check



For $r = 3$, the simplex code has weight enumerator

$$W_{\text{Sim}}(X, Y) = X^7 + 7X^3Y^4.$$

Applying MacWilliams gives

$$W_{\text{Ham}}(X, Y) = X^7 + 7X^4Y^3 + 7X^3Y^4 + Y^7.$$

- ▶ The algebraic transform recovers the geometric weight distribution.
- ▶ Duality is visible both in matrices and in enumerators.

III. Hamming Codes

Q-ary Hamming codes

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Q-ary construction

Choose one nonzero representative from each one-dimensional subspace of $\mathbb{F}_q^r$ and use these vectors as columns of H .

$$n = \frac{q^r - 1}{q - 1}.$$

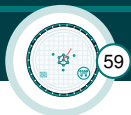
- The resulting code has parameters

$$\left[\frac{q^r - 1}{q - 1}, \frac{q^r - 1}{q - 1} - r, 3 \right]_q.$$

- It is again perfect for one-error correction.

III. Hamming Codes

Why projective representatives



Over $\mathbb{F}_q$, two proportional nonzero columns are linearly dependent.

- ▶ To have $d \geq 3$, no two columns may be proportional.
- ▶ Therefore columns must be projective points, not raw nonzero vectors.
- ▶ Taking all projective points uses the maximum possible number of columns.

Conclusion

Hamming codes are the maximal parity-check configurations with no zero column and no repeated projective point.

III. Hamming Codes

Q-ary perfectness



For the q -ary Hamming code,

$$n = \frac{q^r - 1}{q - 1}, \quad k = n - r.$$

The radius-one sphere has size

$$1 + n(q - 1) = 1 + q^r - 1 = q^r.$$

Hence

$$q^k(1 + n(q - 1)) = q^{n-r} q^r = q^n.$$

Perfect again

Every received word lies at distance at most one from a unique codeword.

III. Hamming Codes

Extended Hamming codes



Extension

Add one overall parity coordinate to force even total weight.

- ▶ The binary $[7, 4, 3]$ code becomes an $[8, 4, 4]$ code.
- ▶ The new minimum distance is 4.
- ▶ It detects three errors and corrects one error.

$$\widehat{\text{Ham}}(3, 2) = [8, 4, 4]_2.$$

III. Hamming Codes

The $[8,4,4]$ code



The extended Hamming code has weight enumerator

$$W(X, Y) = X^8 + 14X^4Y^4 + Y^8.$$

- ▶ All codeword weights are divisible by 4.
- ▶ The code is self-dual.
- ▶ It is the smallest doubly-even self-dual binary code.

Connection

This code is the length-8 seed behind the E_8 lattice construction.

III. Hamming Codes

Automorphisms



Symmetry group

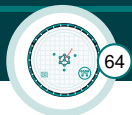
The automorphism group of the binary Hamming code is induced by invertible linear transformations of $\mathbb{F}_2^r$:

$$\text{Aut}(\text{Ham}(r, 2)) \cong \text{GL}(r, 2).$$

- ▶ A matrix in $\text{GL}(r, 2)$ permutes the nonzero columns of H_r .
- ▶ Hence it permutes coordinates and preserves the kernel.
- ▶ The code remembers the projective geometry.

III. Hamming Codes

Tanner graph



- ▶ The Tanner graph of H_r has $n = 2^r - 1$ variable nodes and r check nodes.
- ▶ It is dense compared with modern LDPC codes.
- ▶ But it has an exceptionally simple exact decoder.

Interpretation

For Hamming codes, the syndrome is not iteratively refined; it is read directly as a projective point.

III. Hamming Codes

Failure modes



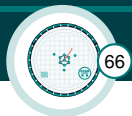
- ▶ A two-error pattern has syndrome equal to the sum of two columns.
- ▶ In a binary Hamming matrix, that sum is another nonzero column.
- ▶ Therefore a two-error pattern may be miscorrected as a one-error pattern.

Moral

Minimum distance 3 guarantees single-error correction, not reliable detection after attempting correction.

III. Hamming Codes

SECDED



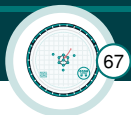
SECDED

Single-error correction, double-error detection is obtained by using the extended Hamming code.

- ▶ The extension raises the distance from 3 to 4.
- ▶ One error is corrected by the Hamming syndrome.
- ▶ Two errors are detected because parity and syndrome disagree in a recognizable way.

III. Hamming Codes

Shortened Hamming codes



Shortening

Shortened Hamming codes are obtained by fixing selected coordinates to zero and deleting them.

- ▶ They are useful when exact lengths $2^r - 1$ are inconvenient.
- ▶ The minimum distance remains at least 3.
- ▶ Perfectness is usually lost.

Engineering

Many practical memory codes are shortened or extended Hamming codes.

III. Hamming Codes

Hamming codes as BCH codes



Cyclic viewpoint

When $n = 2^r - 1$, the binary Hamming code can be realized as a narrow-sense BCH code of designed distance 3.

- ▶ Choose a primitive n -th root $\alpha \in \mathbb{F}_{2^r}$.
- ▶ Let $g(X)$ be the minimal polynomial of α over $\mathbb{F}_2$.
- ▶ Then the cyclic code generated by $g(X)$ is a Hamming code.

III. Hamming Codes

The $[7,4,3]$ cyclic form



Over $\mathbb{F}_2$,

$$X^7 - 1 = (X + 1)(X^3 + X + 1)(X^3 + X^2 + 1).$$

Taking

$$g(X) = X^3 + X + 1$$

gives a cyclic $[7, 4, 3]$ Hamming code.

- ▶ This is the bridge from projective checks to polynomial ideals.
- ▶ The next chapter makes that bridge systematic.

III. Hamming Codes

Classification intuition



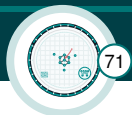
- ▶ Perfect one-error-correcting linear codes are essentially Hamming codes, apart from trivial cases.
- ▶ The parity-check matrix must use every possible nonzero syndrome once.
- ▶ Thus the construction is forced by the Hamming bound.

Mathematical taste

Hamming codes are beautiful because optimality, decoding, and geometry are the same statement.

III. Hamming Codes

Exercise: distance



Problem

Let H contain one representative of every point of $\mathbb{P}^{r-1}(\mathbb{F}_q)$. Prove that $\ker H$ has distance exactly 3.

Hint.

- ▶ No one or two columns are dependent.
- ▶ Any projective line contains three dependent representatives after scaling.
- ▶ Translate column dependence into a codeword support.

III. Hamming Codes

Exercise: perfectness

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Problem

Verify directly that the q -ary Hamming code meets the Hamming bound with equality for $t = 1$.

$$n = \frac{q^r - 1}{q - 1}, \quad k = n - r.$$

Computation.

$$q^k(1 + n(q - 1)) = q^{n-r}(1 + q^r - 1) = q^n.$$

III. Hamming Codes

Exercise: dual weights



Problem

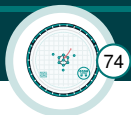
Show that every nonzero word of the binary simplex code has weight 2^{r-1} .

Hint. For a nonzero row vector $a \in \mathbb{F}_2^r$, the word aH_r evaluates the nonzero linear functional $x \mapsto ax^T$ on every nonzero vector x .

- ▶ Exactly half of all vectors of $\mathbb{F}_2^r$ evaluate to 1.
- ▶ The zero vector evaluates to 0.

III. Hamming Codes

Summary



Construction

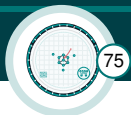
- ▶ parity-check columns are projective points
- ▶ no repeated projective column
- ▶ all syndromes are used

Consequences

- ▶ parameters $[2^r - 1, 2^r - 1 - r, 3]_2$
- ▶ perfect one-error correction
- ▶ dual simplex code

III. Hamming Codes

Transition to cyclic codes



New symmetry

Hamming codes are governed by projective geometry. Cyclic codes are governed by the cyclic shift operator.

- ▶ Linear code: subspace of $\mathbb{F}_q^n$.
- ▶ Hamming code: subspace defined by projective parity checks.
- ▶ Cyclic code: subspace stable under a cyclic permutation.

shift symmetry $\iff$ ideals in $\mathbb{F}_q[X]/(X^n - 1)$.

IV. Cyclic Codes

IV. Cyclic Codes

Cyclic shift



Cyclic code

A linear code $\mathcal{C} \leq \mathbb{F}_q^n$ is cyclic if

$$(c_0, c_1, \dots, c_{n-1}) \in \mathcal{C} \quad \Rightarrow \quad (c_{n-1}, c_0, \dots, c_{n-2}) \in \mathcal{C}.$$

- ▶ This is invariance under a single permutation.
- ▶ Since the permutation has order n , cyclic codes are modules over a cyclic group algebra.
- ▶ The polynomial model makes this precise.

IV. Cyclic Codes

Polynomial dictionary



Vector to polynomial

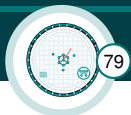
Identify

$$(c_0, c_1, \dots, c_{n-1}) \longleftrightarrow c(X) = c_0 + c_1X + \dots + c_{n-1}X^{n-1}.$$

- ▶ Multiplication by X shifts coefficients.
- ▶ The relation $X^n = 1$ performs wrap-around.
- ▶ Therefore we work modulo $X^n - 1$.

IV. Cyclic Codes

The quotient ring



Ambient ring

$$R_n = \mathbb{F}_q[X]/(X^n - 1).$$

- ▶ Vectors of length n are polynomials of degree $< n$.
- ▶ Cyclic shift is multiplication by X in R_n .
- ▶ A cyclic code is an $\mathbb{F}_q$ -subspace stable under multiplication by X .

Key question

Which subspaces of R_n are stable under multiplication by X ?

IV. Cyclic Codes

Ideals



Cyclic codes are ideals

Under the identification $\mathbb{F}_q^n \cong R_n$, cyclic codes are exactly the ideals of R_n .

Proof.

If $\mathcal{C}$ is stable under multiplication by X , then it is stable under multiplication by every polynomial in X . Hence it is an ideal. The converse is immediate. $\square$

Consequence

The classification of cyclic codes is an ideal-theoretic problem.

IV. Cyclic Codes

Principal ideals



Principal generator

Every ideal of R_n is generated by a unique monic divisor $g(X)$ of $X^n - 1$:

$$\mathcal{C} = (g(X)).$$

- ▶ This follows from the principal ideal property of $\mathbb{F}_q[X]$.
- ▶ The generator polynomial is the monic element of least degree in the code.
- ▶ The divisibility condition is $g(X) \mid X^n - 1$.

IV. Cyclic Codes

Dimension

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Dimension formula

If $\mathcal{C} = (g(X)) \subseteq R_n$ and $\deg g = r$, then

$$\dim_{\mathbb{F}_q} \mathcal{C} = n - r.$$

Proof.

The polynomials

$$g(X), Xg(X), \dots, X^{n-r-1}g(X)$$

form an $\mathbb{F}_q$ -basis of the ideal. □

IV. Cyclic Codes

Generator matrix from shifts



If

$$g(X) = g_0 + g_1X + \cdots + g_rX^r,$$

then a generator matrix is obtained by shifting the coefficient vector:

$$\begin{bmatrix} g_0 & g_1 & \cdots & g_r & 0 & \cdots & 0 \\ 0 & g_0 & g_1 & \cdots & g_r & \cdots & 0 \\ \vdots & & \ddots & & & \ddots & \vdots \end{bmatrix}.$$

- ▶ This matrix is Toeplitz-like.
- ▶ It encodes multiplication by $g(X)$.

IV. Cyclic Codes

Parity-check polynomial

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Check polynomial

If $g(X) \mid X^n - 1$, define

$$h(X) = \frac{X^n - 1}{g(X)}.$$

- ▶ $\deg h = k$ when $\mathcal{C}$ has dimension k .
- ▶ Since $g(X)h(X) = 0$ in R_n , multiplication by $h(X)$ checks membership.
- ▶ More precisely, $c(X) \in (g)$ implies $c(X)h(X) \equiv 0 \pmod{X^n - 1}$.

IV. Cyclic Codes

Reciprocal polynomials



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Reciprocal

For $f(X) = f_0 + \cdots + f_m X^m$ with $f_0 \neq 0$, define

$$f^*(X) = X^m f(X^{-1}).$$

- ▶ Parity-check matrices are naturally written using reciprocal polynomials.
- ▶ The dual of a cyclic code is cyclic.
- ▶ If $\mathcal{C} = (g)$ and $h = (X^n - 1)/g$, then $\mathcal{C}^\perp$ is generated by a reciprocal of h .

IV. Cyclic Codes

Idempotents



Idempotent generator

When $\gcd(n, q) = 1$, each cyclic code has a unique idempotent generator $e(X)$ satisfying

$$e(X)^2 = e(X), \quad \mathcal{C} = (e(X)).$$

- ▶ Idempotents decompose the semisimple ring R_n .
- ▶ They expose the representation-theoretic structure.
- ▶ Generator polynomials are usually better for computation.

IV. Cyclic Codes

Roots



Let α be an n -th root of unity in an extension field.

▶ A cyclic code generated by $g(X)$ has zeros α^i where $g(\alpha^i) = 0$.

▶ The defining set is

$$Z(\mathcal{C}) = \{i \pmod n : g(\alpha^i) = 0\}.$$

▶ Root sets translate algebraic divisibility into combinatorics modulo n .

IV. Cyclic Codes

Cyclotomic cosets



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Q-cyclotomic coset

Modulo n , the q -cyclotomic coset of i is

$$C_i = \{i, iq, iq^2, \dots\} \pmod{n}.$$

- ▶ If α^i is a root over $\mathbb{F}_q$, all conjugates α^{iq^j} are roots.
- ▶ Therefore defining sets are unions of cyclotomic cosets.
- ▶ Minimal polynomials correspond to these cosets.

IV. Cyclic Codes

Minimal polynomials



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Minimal polynomial

For α^j , the minimal polynomial over $\mathbb{F}_q$ is

$$M_i(X) = \prod_{j \in \mathcal{C}_i} (X - \alpha^j).$$

- ▶ It lies in $\mathbb{F}_q[X]$.
- ▶ It is irreducible.
- ▶ Generator polynomials of cyclic codes are products of such $M_i(X)$.

IV. Cyclic Codes

BCH codes

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BCH design

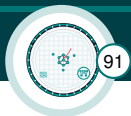
A BCH code is a cyclic code whose generator polynomial has a prescribed block of consecutive roots.

$$g(\alpha^b) = g(\alpha^{b+1}) = \dots = g(\alpha^{b+\delta-2}) = 0.$$

- ▶ The integer δ is the designed distance.
- ▶ Consecutive roots force many syndromes to vanish.
- ▶ The BCH bound converts this into a distance guarantee.

IV. Cyclic Codes

BCH bound



BCH bound

If a cyclic code has $\delta - 1$ consecutive roots

$$\alpha^b, \alpha^{b+1}, \dots, \alpha^{b+\delta-2},$$

then its minimum distance is at least δ .

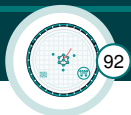
Proof.

A codeword of weight $< \delta$ would give a nonzero polynomial with too many consecutive zero syndromes, forcing a Vandermonde determinant to vanish.



IV. Cyclic Codes

Narrow-sense primitive BCH



Common special case

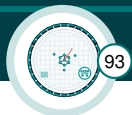
Let $n = q^m - 1$ and let α be primitive in $\mathbb{F}_{q^m}$. The narrow-sense BCH code of designed distance δ has roots

$$\alpha, \alpha^2, \dots, \alpha^{\delta-1}.$$

- ▶ Primitive: the length is the full multiplicative order.
- ▶ Narrow-sense: the first root is α .
- ▶ Hamming codes occur when $\delta = 3$ in the binary case.

IV. Cyclic Codes

Cyclic Hamming code



Let $n = 2^r - 1$ and let $\alpha \in \mathbb{F}_{2^r}$ be primitive.

- ▶ Let $g(X)$ be the minimal polynomial of α over $\mathbb{F}_2$.
- ▶ Then $\deg g = r$.
- ▶ The cyclic code (g) has parameters

$$[2^r - 1, 2^r - 1 - r, 3]_2.$$

Identification

This is the binary Hamming code in cyclic coordinates.

IV. Cyclic Codes

The case n equals 7



Over $\mathbb{F}_2$,

$$X^7 - 1 = (X + 1)(X^3 + X + 1)(X^3 + X^2 + 1).$$

- ▶ A primitive 7-th root has minimal polynomial $X^3 + X + 1$ or $X^3 + X^2 + 1$.
- ▶ Either cubic gives a cyclic $[7, 4, 3]$ Hamming code.
- ▶ The two choices are equivalent under coordinate reversal.

IV. Cyclic Codes

Encoding by multiplication



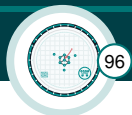
For a cyclic code (g) with $\deg g = r$, non-systematic encoding is

$$m(X) \mapsto c(X) = m(X)g(X), \quad \deg m < n - r.$$

- ▶ The message polynomial has $k = n - r$ coefficients.
- ▶ The product automatically lies in the ideal (g) .
- ▶ Hardware can implement this multiplication by shift registers.

IV. Cyclic Codes

Systematic cyclic encoding



Let $\deg g = r$. To encode a message $m(X)$ with $\deg m < k$:

$$X^r m(X) = q(X)g(X) + p(X), \quad \deg p < r.$$

Then set

$$c(X) = X^r m(X) - p(X).$$

- ▶ Since $c(X)$ is divisible by $g(X)$, it is a codeword.
- ▶ The high-degree coefficients contain the message.
- ▶ The low-degree polynomial $p(X)$ supplies parity.

IV. Cyclic Codes

Syndrome polynomial



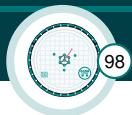
For received $v(X) = c(X) + e(X)$, compute the remainder

$$s(X) = v(X) \bmod g(X).$$

- ▶ Since $c(X)$ is divisible by $g(X)$, $s(X) = e(X) \bmod g(X)$.
- ▶ This is the polynomial analogue of $Hv^T = He^T$.
- ▶ BCH decoding refines this by evaluating at roots of g .

IV. Cyclic Codes

Root syndromes



If α^i is a root of $g(X)$, then every codeword satisfies

$$c(\alpha^i) = 0.$$

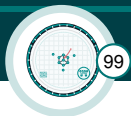
For $v = c + e$,

$$v(\alpha^i) = e(\alpha^i).$$

- ▶ The numbers $S_i = v(\alpha^i)$ are syndromes.
- ▶ Decoding asks for the error locations and values from these sums.
- ▶ This is a finite-field moment problem.

IV. Cyclic Codes

Error locator polynomial



For error locations $j_1, \dots, j_t$, define

$$\Lambda(Z) = \prod_{\ell=1}^t (1 - \alpha^{j_\ell} Z).$$

- ▶ The roots of Λ identify error locations.
- ▶ BCH and Reed–Solomon decoding compute Λ from syndromes.
- ▶ Berlekamp–Massey is an efficient algorithm for this recurrence problem.

IV. Cyclic Codes

LFSR viewpoint



Linear feedback shift register

Division by $g(X)$ can be implemented with a shift register whose feedback taps are the nonzero coefficients of $g(X)$.

- ▶ Memory cost is $\deg g = n - k$ registers.
- ▶ Input is processed one symbol per clock.
- ▶ This is why cyclic codes became central in hardware error control.

IV. Cyclic Codes

Burst errors



Burst

A burst of length ℓ is an error whose nonzero positions lie in ℓ consecutive coordinates.

- ▶ Polynomial remainders are well suited for burst detection.
- ▶ CRC polynomials are chosen to detect common burst patterns.
- ▶ This is detection rather than correction in many communication protocols.

IV. Cyclic Codes

CRC codes



CRC uses a generator polynomial $g(X)$ and transmits a multiple of $g(X)$.

- ▶ Receiver divides by $g(X)$.
- ▶ Nonzero remainder means an error was detected.
- ▶ The method is a cyclic-code membership test.

Mathematical core

CRC is not a different theory: it is cyclic codes used primarily for error detection.

IV. Cyclic Codes

Reed–Solomon codes



Reed–Solomon

Over $\mathbb{F}_q$, Reed–Solomon codes evaluate polynomials of degree $< k$ at n distinct field points.

- ▶ In a suitable form, they are cyclic codes.
- ▶ They meet the Singleton bound: $d = n - k + 1$.
- ▶ They correct symbol errors, which is why they are strong against bursts.

IV. Cyclic Codes

BCH versus Reed–Solomon



BCH

- ▶ often binary
- ▶ roots in an extension field
- ▶ designed distance
- ▶ strong for bit errors

Reed–Solomon

- ▶ nonbinary
- ▶ evaluation over $\mathbb{F}_q$
- ▶ MDS property
- ▶ strong for symbol bursts

IV. Cyclic Codes

Dual cyclic codes



Dual cyclicity

The dual of a cyclic code is cyclic.

Proof.

If c is orthogonal to every word of $\mathcal{C}$, then a cyclic shift of c is also orthogonal to every cyclic shift of every word of $\mathcal{C}$. Since $\mathcal{C}$ is shift-invariant, the shifted c lies in $\mathcal{C}^\perp$. □

- ▶ Generator polynomials of duals are expressed using reciprocal polynomials.
- ▶ This is another sign that cyclic codes form a closed algebraic world.

IV. Cyclic Codes

Repeated-root ~~caution~~



Assumption often made

Many clean statements assume $\gcd(n, q) = 1$.

- ▶ Then $X^n - 1$ has no repeated roots.
- ▶ The quotient ring is semisimple.
- ▶ If $\gcd(n, q) \neq 1$, repeated-root cyclic codes require extra care.

Moral

Always check separability before using root-set arguments naively.

IV. Cyclic Codes

Quasi-cyclic hint



Quasi-cyclic code

A code is quasi-cyclic of index ℓ if it is invariant under cyclic shift by ℓ positions.

- ▶ Generator matrices are built from circulant blocks.
- ▶ Circulant blocks correspond to polynomials modulo $X^m - 1$.
- ▶ This structure is important in code-based cryptography.

IV. Cyclic Codes

Algebraic hierarchy



linear code
↓ add cyclic shift
cyclic code
↓ prescribe roots
BCH code
↓ evaluation/MDS structure
Reed–Solomon code

Perspective

Each additional algebraic structure buys stronger theorems and faster algorithms.

IV. Cyclic Codes

Cyclic codes as modules



- ▶ A linear code is an $\mathbb{F}_q$ -subspace of $\mathbb{F}_q^n$.
- ▶ A cyclic code is an $\mathbb{F}_q[X]$ -submodule, where X acts by shift.
- ▶ The relation $X^n = 1$ makes the module a module over R_n .

cyclic code = submodule of R_n = ideal of R_n .

Mathematician's slogan

Shift-invariant linear algebra is ideal theory.

IV. Cyclic Codes

Exercise: ideal proof



Problem

Prove that if an $\mathbb{F}_q$ -subspace $\mathcal{C} \subseteq R_n$ is closed under multiplication by X , then it is an ideal.

Hint.

- ▶ Closure under X gives closure under X^j for every $j \geq 0$.
- ▶ By linearity, closure under every polynomial follows.
- ▶ Reduce modulo $X^n - 1$.

IV. Cyclic Codes

Exercise: dimension



Problem

Let $g(X) \mid X^n - 1$ and $\deg g = r$. Show that (g) has dimension $n - r$ as an $\mathbb{F}_q$ -space.

Hint.

$$a(X)g(X), \quad \deg a < n - r$$

are all distinct modulo $X^n - 1$.

- ▶ Count the possible coefficients of $a(X)$.
- ▶ Use the degree bound to prove injectivity.

IV. Cyclic Codes

Exercise: cyclic Hamming



Problem

For $X^7 - 1 = (X + 1)(X^3 + X + 1)(X^3 + X^2 + 1)$ over $\mathbb{F}_2$, show that the code generated by $g = X^3 + X + 1$ has parameters $[7, 4, 3]$.

Hint.

- ▶ Dimension is $7 - \deg g = 4$.
- ▶ The BCH bound with roots α, α^2 gives $d \geq 3$.
- ▶ The Singleton bound excludes $d > 4$, and a direct word gives $d = 3$.

IV. Cyclic Codes

Summary of cyclic codes



Algebra

- ▶ $R_n = \mathbb{F}_q[X]/(X^n - 1)$
- ▶ cyclic codes are ideals
- ▶ generator polynomial g
- ▶ root sets and cosets

Algorithms

- ▶ encoding by multiplication
- ▶ parity by remainders
- ▶ BCH decoding from syndromes
- ▶ LFSR implementation

IV. Cyclic Codes

Final perspective



Linear codes turn error correction into finite-dimensional algebra.

Hamming codes show how perfect decoding is finite geometry.

Cyclic codes show how shift symmetry becomes polynomial ideal theory.

Coding theory is the art of adding structure to redundancy.